

Clustering mixed-length shapelets (subsequences of varying lengths from time series) is a key challenge in time series analysis, especially for unsupervised discovery and clustering of discriminative patterns (often called **U-Shapelets** or unsupervised shapelets).

What are Shapelets?

Shapelets are short, discriminative **subsequences** (local patterns) in time series data. Originally from supervised classification, they were extended to unsupervised settings for clustering.

- In **supervised** use: Shapelets maximize class separation.
- In **unsupervised** use (U-Shapelets): Shapelets maximize separation between groups of time series based on how well a subsequence matches (via minimum distance) across the dataset.

The Mixed-Length Challenge

Shapelets naturally have **different lengths** because meaningful patterns (e.g., a short spike vs. a longer oscillation) vary in duration. Direct clustering of raw shapelets is difficult because:

- Euclidean distance (or similar) requires **equal lengths**.
- Longer shapelets tend to dominate similarity measures without normalization.
- There are huge numbers of candidates (all possible subsequences across many lengths).

Main Approaches to Handle Mixed-Length Shapelets

Here are the primary strategies:

1. U-Shapelets (Unsupervised Shapelets) — Classic Approach (Zakaria et al. 2012, Ulanova et al. 2015)

- **Core Idea:** Don't cluster the shapelets themselves directly . Instead , find discriminative subsequences (U-Shapelets) of various lengths , then use their **distance vectors** to the full time series for clustering .
- **Process**
 - Generate candidate subsequences across a range of lengths .
 - For each candidate s of length l , compute the **subsequence distance** (s_{dist}) to every time series T (minimum distance to any subsequence of T of length l).
 - Create an **orderline** (sorted list of distances).
 - Score candidates using a **gap statistic** (separation between two groups based on mean/std of distances).
 - Iteratively select good U-Shapelets and remove matched time series.
- **Handles mixed lengths** naturally because each candidate is evaluated independently with its own length .
- **Scalability fix** (SUS algorithm): Use discretization (e.g., SAX) + hashing to avoid checking millions of candidates .

2. Learning-Based Methods (Modern)

- **AUTOSHAPE** (Li et al. 2022): Uses an autoencoder to learn embeddings for variable-length shapelet candidates . Self-supervised loss unifies representations across lengths + diversity loss for discriminative ones . Then cluster in the embedding space.
- **Contrastive Shapelet Learning (CSL)**: Learns shapelets of multiple scales/lengths jointly , then transforms time series into feature vectors for standard clustering (k-means, HDBSCAN etc.).

- **Multi-scale / Multi-view:** Explicitly maintain separate shapelet sets for different lengths (scales) and combine them.

3. Embedding + Standard Clustering

- Extract many mixed-length shapelets.
- Transform each time series into a **feature vector** (e.g., distances to the top k shapelets, or occurrence counts/positions).
- Cluster the time series in this new space using k-means, hierarchical clustering, DBSCAN, etc.
- This indirectly clusters the underlying patterns via their effect on the data.

4. Direct Shapelet Clustering (Less Common)

- Normalize shapelets (z-normalize + length normalization).
- Use length-invariant distances (e.g., normalized cross-correlation) or pad/resample (risky).
- Cluster with hierarchical methods or k-medoids.
- Usually inferior to the "find discriminative ones first" paradigm.

Practical Workflow (Recommended)

1. **Candidate Generation**: Slide windows of lengths from min_l to max_l (e.g., 5% to 50% of series length).
2. **Quality Evaluation**:
 - Use gap score (U-Shapelets).
 - Or use contrastive /self-supervised objectives.
3. **Selection**: Keep diverse, high-quality shapelets (avoid redundant /self-similar ones).
4. **Transformation**: Build distance matrix or feature matrix from selected

shapelets .

5. Clustering:

- Cluster the *time series* (most common).
- Or cluster the shapelets if you want pattern groups (e.g., "all rising edges", "all plateaus ").

Tools & Libraries

- **aeon** (Python): Strong shapelet support .
- **tslearn**: LearningShapelets .
- **UCR Archive** implementations for U-Shapelets.
- Custom : Many papers release code (e.g., AUTOSHAPEon arXiv).

Key Advantages of Mixed-Length Handling

- Better captures real patterns (short bursts + longer trends).
- Works with variable-length time series datasets .
- Highly interpretable (you can visualize the representative shapelets per cluster).

Would you like a deeper dive into any specific method (U-Shapelets algorithm details , code example , or comparison), or are you working with a particular dataset /type of time series?